EXP 4:

Program:

```
# Build an Artificial Neural Network using Backpropagation

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, classification_report

# Load dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create ANN model
ann = MLPClassifier(
    hidden_layer_sizes=(5,),
    activation='relu',
    solver='adam',
    max_iter=1000,
    random_state=42
)
```

```
# Train the model (uses Backpropagation)
ann.fit(X_train, y_train)

# Predict test data
y_pred = ann.predict(X_test)

# Display results
print("Actual Labels:")
print(y_test)

print("\nPredicted Labels:")
print(y_pred)

print("\nAccuracy:")
print(accuracy_score(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

OUTPUT:

```python
# Train the model (uses Backpropagation)
ann.fit(X_train, y_train)

# Predict test data
y_pred = ann.predict(X_test)

# Display results
print("Actual Labels:")
print(y_test)

print("\nPredicted Labels:")
print(y_pred)

print("\nAccuracy:")
print(accuracy_score(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

```
Actual Labels:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Predicted Labels:
[1 0 2 1 2 0 1 2 2 1 2 0 0 0 0 1 2 1 2 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Accuracy:
0.9333333333333333

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        19
           1       1.00      0.77      0.87        13
           2       0.81      1.00      0.90        13

    accuracy                           0.93        45
   macro avg       0.94      0.92      0.92        45
weighted avg       0.95      0.93      0.93        45
```